Problem

At my karate dojo, we don't have a reliable way to track attendance, and it was mostly manually done by parents. However, this process was often inconsistent as parents forgot to update the sheet periodically. Creating a reliable system for attendance would allow for accurate attendance to use later for analyzing training consistency and connecting attendance to tournament performance.

Solution

Through my device, each student is given their own NFC tag. When they arrive, they simply tap their tag on the scanner.

<u>Code process:</u>

1. The PN532 NFC reader scans the tag
2. The ESP32 reads the tag's unique ID and matches it to a student name
3. The ESP32 connects to WiFi and sends a request to a Google Apps Script web app
4. The script updates a Google Sheet and marks the student as present for that day
5. LEDs on the device give instant feedback indicating success or failure

<u>Features</u>

- Fast "tap to check-in" system
- Automatically logs attendance by date
- Initializes attendance rows and fills missing students as absent
- Real-time google sheet updates

Improvements (Based on User Feedback)

After testing the system at home, I noticed that some features for the user experience weren't clear.

Specific improvements:

- Added a quick LED flash sequence right after scanning to show the tag was detected
- Made the red LED stay on while the system is processing (waiting for the response from the server)
- Added a green LED confirmation when attendance is successfully marked

- Added a red LED error signal if something fails

Overall benefits

I have been deploying the system for a week and here are the improvements I have noticed for attendance:

- System is accurate and there aren't incorrect check-ins
- Checkins take seconds to register
- Attendance provides useful data can be analyzed over time
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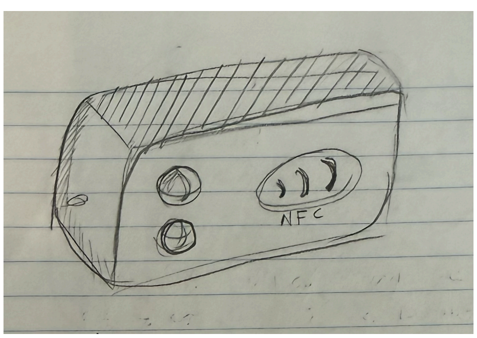
Future Improvements

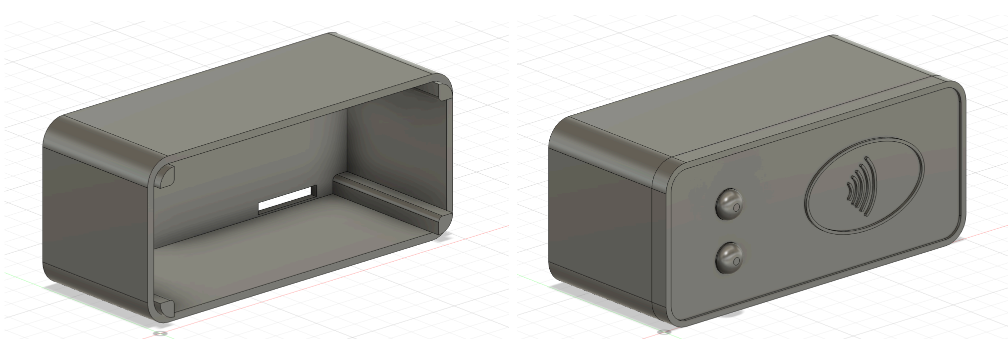
- Online app to analyze attendance data
- More advanced stats through linking attendance with performance or other IOT devices
- Wireless or Offline modes

Images and Deployment:

Initial design:



3D CAD Model:



Final Product:

[VIDEO LINK]